HOW TO CREATE A DOCKER PROJECT FOLDER IN VS CODE

Step-by-step practical guide

Objective: Create a folder named my-first-docker-project containing index.html, style.css, and Dockerfile. These files will be used to build and run a simple website with Docker Desktop.

Step 1: Create the project folder

1. Go to your computer's Desktop or Documents folder.
2. Right-click an empty space and select New > Folder.
3. Name the folder my-first-docker-project.

Step 2: Open the folder in Visual Studio Code

4. Open Visual Studio Code.
5. Select File > Open Folder.
6. Locate and select my-first-docker-project.
7. Click Select Folder. The folder should appear in the Explorer panel.

Step 3: Create index.html

In the Explorer panel, right-click the project folder, select New File, name it index.html, and enter this code. Save with Ctrl + S.

```
<!DOCTYPE html>
<html>
<head>
    <title>My Docker Application</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <h1>Welcome to Docker!</h1>
    <p>My first application is running
       inside a Docker container.</p>
</body>
</html>
```

Step 4: Create style.css

Right-click the project folder, select New File, name it style.css, enter the code below, and save.

```
body {
    font-family: Arial, sans-serif;
    text-align: center;
    background: #eaf2f8;
    padding: 50px;
}
h1 {
    color: #1976d2;
}
```

Step 5: Create the Dockerfile

Create a new file named exactly Dockerfile, without .txt or another extension. Enter the following instructions and save.

```
# Use Nginx as the web server
FROM nginx:alpine

# Copy website files into the Nginx directory
COPY . /usr/share/nginx/html

# Document the container web port
EXPOSE 80
```

Step 6: Verify the project structure

The VS Code Explorer should show:

```
my-first-docker-project/
│
├── index.html
├── style.css
└── Dockerfile
```

Step 7: Build the Docker image

Select Terminal > New Terminal in VS Code. Make sure the terminal is inside the project folder, then run:

```
docker build -t my-first-app .
```

This builds an image named my-first-app using the Dockerfile in the current folder. The first build may take time while Docker downloads the Nginx base image.

Step 8: Run the Docker container

Execute:

```
docker run -d -p 8080:80 --name my-first-container my-first-app
```

The command runs the container in the background and maps computer port 8080 to container port 80.

Step 9: Verify and open the application

Check the running container:

```
docker ps
```

Open a browser and visit:

```
http://localhost:8080
```

You should see the heading Welcome to Docker! and the paragraph from your website.

Alternative: Create the folder and empty files using PowerShell

You can create the folder and empty files using the VS Code PowerShell terminal:

```
mkdir my-first-docker-project
cd my-first-docker-project
New-Item index.html -ItemType File
New-Item style.css -ItemType File
New-Item Dockerfile -ItemType File
code .
```

Then open each file, paste the corresponding code from Steps 3–5, and save.

Useful reminders

- The Dockerfile must be in the project root, alongside index.html and style.css.
- Check that the file is named Dockerfile, not Dockerfile.txt. In Windows File Explorer, enable View > Show > File name extensions.
- Run docker build from the folder containing the Dockerfile.
- If port 8080 is already in use, use -p 8081:80 and visit http://localhost:8081.